Practice Questions

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Score: 86% Passing Score: 80%

Question 1: ✔ Correct

You need to perform a reverse lookup of the 10.0.0.3 IP address. Which command can you use to accomplish this? (Select two. Each response is a complete solution.)

- [ ] nbtstat -a 10.0.0.3
- [x] nslookup 10.0.0.3
- [ ] arp 10.0.0.3
- [ ] ipconfig /dnslookup 10.0.0.3
- [x] dig -x 10.0.0.3

EXPLANATION

To perform a reverse lookup of the 10.0.0.3 IP address, use either of the following commands:

- **dig -x 10.0.0.3**
- **nslookup 10.0.0.3**

The **ipconfig** command is used to perform a forward or reverse DNS lookup. The **arp 10.0.0.3** command displays the MAC address of the network host with an IP address of 10.0.0.3. The **nbtstat -a 10.0.0.3** command displays the NETBIOS name of the host assigned an IP address of 10.0.0.3.

REFERENCES

LabSim for Network Pro, Section 5.10.

Question 2: ✔ Correct

Mary calls to tell you that she can't connect to an intranet server called WebSrv1. From her computer, you ping the server's IP address. The ping test is successful. Which tool would you use on her workstation next to troubleshoot the problem?

- ➡ **nslookup**
- **netstat**
- **tracert**
- **arp**
- **nbtstat**

EXPLANATION

Use **nslookup** to troubleshoot name resolution problems. Because the ping test was successful, you know that both the client and the server can communicate using TCP/IP with IP addresses. This tells you that the problem is related to name resolution.

REFERENCES

LabSim for Network Pro, Section 5.10.

Question 3: ✔ Correct

You are troubleshooting a network connectivity issue on a Unix system. You are able to connect to remote systems by using their IP address, but unable to connect using the host name. You check the TCP/IP configuration and note that a DNS server IP address is configured.

You decide to run some manual resolution queries to ensure that the communication between the Unix system and the DNS server are working correctly. Which utilities can you use to do this? (Choose two.)

- ➡ ☑ **dig**
- ➡ ☑ **nslookup**
- ☐ **traceroute**
- ☐ **tracert**

EXPLANATION

The **dig** and **nslookup** commands allow you to perform manual DNS lookups from a Linux or Unix system. This can be very useful when you are troubleshooting name resolution issues.

Use **tracert** and **traceroute** to track the route that a packet takes as it crosses a network. You would not typically use these commands to troubleshoot a name resolution problem, though they may be useful if you are unable to connect to the DNS server.

REFERENCES

LabSim for Network Pro, Section 5.10.

Question 4: ✔ Correct

Examine the following output:

```
Server: to.xct.mirrorxhq.net
Address: 209.53.4.130
Name: westxsim.com
Address: 64.78.193.84
```

Which of the following utilities produced this output?

- ipconfig
- ➡ **nslookup**
- netstat
- tracert

EXPLANATION

The output is from the **nslookup** command on a Windows Server system. **nslookup** is a tool that allows you to send manual DNS resolution requests to a DNS server. The output displays the IP address and host name of the DNS server that performed the resolution, and the IP address and host name of the target specified for resolution. **nslookup** can be a useful tool when troubleshooting DNS name resolution problems.

The **ipconfig** utility is used on a Windows system to view the TCP/IP configuration of network interfaces. **netstat** is used to view protocol connections that have been established by the system, as well as what incoming TCP/IP ports are in use by the system. **tracert** is a tool used to view information on the route a packet takes as it traverses the network to a remote host.

REFERENCES

LabSim for Network Pro, Section 5.10.

Question 5: ✕ Incorrect

Consider the following output.

```
;; res options: init recurs defnam dnsrch
;;got answer:
;;->>HEADER<<-opcode:QUERY, status; NOERROR,id:4
;;flags: qr rd ra; QUERY:1, ANSWER:1, AUTHORITY:2, ADDITIONAL:0
;;QUERY SECTION:
;; westsim111.com, type = A, class = IN

;;ANSWER SECTION:
westsim111.com. 7h33m IN A 76.141.43.129
;;AUTHORITY SECTION:
westsim111.com. 7h33m IN NS dns1.deriatct111.com.
westsim111.com. 7h33m IN NS dns2.deriatct222.com.
;;Total query time: 78 msec
;;FROM: localhost.localdomain to SERVER: default -- 202.64.49.150
;;WHEN: Tue Feb 16 23:21:24 2005
;;MSG SIZE sent: 30 rcvd: 103
```

Which of the following utilities produced this output?

- **nbtstat**
- ➡ **dig**
- **ping**
- **nslookup**

EXPLANATION

The output shown is from the **dig** command run on a Linux system. Although **nslookup** and **dig** provide some of the same information, you can tell this output came from **dig** because **dig** produces significantly more detail in its default usage.

Use **nbtstat** to view information on the NetBIOS over TCP/IP (NetBT) name resolutions that have been performed. Use **ping** to test connectivity between systems on a network.

REFERENCES

LabSim for Network Pro, Section 5.10.

Question 6: ✔ Correct

Consider the following output from a **dig** command run on a Linux system.

```
; <<>> DiG 8.2 <<>> westsim111.com
;;res options:init recurs defnam dnsrch
;;got answer:
;;->>HEADER<<-opcode:QUERY, status: NOERROR, id:4
;;flags: qr rd ra; QUERY:1, ANSWER:1, AUTHORITY:2, ADDITIONAL:0
;;QUERY SECTION:
;; westsim111.com, type = A, class = IN

;;ANSWER SECTION:
westsim111.com. 7h33m IN A 76.141.43.129
;;AUTHORITY SECTION:
westsim111.com. 7h33m IN NS dns1.deriatct111.com.
westsim111.com. 7h33m IN NS dns2.deriatct222.com.
;;Total query time: 78 msec
;;FROM: localhost.localdomain to SERVER:default -- 202.64.49.150
;;WHEN: Tue Feb 16 23:21:24 2005
;;MSG SIZE sent: 30 rcvd:103
```

What is the IP address of the DNS server that performed this name resolution?

- 192.168.1.100
- 76.141.43.129
- 16.23.21.24
- ➡ 202.64.49.150

EXPLANATION

When the **dig** command is used to perform a manual DNS lookup, a range of information is provided. The IP address of the DNS server that performed the name resolution is shown in the bottom area of the output, on the end of the **;;FROM** line.

The IP address shown in the answer section denotes the resolved IP address for the domain or host for which the resolution was requested. In this case, that address is 76.141.43.129.

The other two answers are invalid.

REFERENCES

LabSim for Network Pro, Section 5.10.

Question 7: ✔ Correct

A user reports that he can't browse to a specific website on the internet.

From his computer, you find that a ping test to the web server succeeds. A traceroute test shows 17 hops to the destination web server.

What is the most likely cause of the problem?

- Incorrect default gateway address
- Incorrect subnet mask value
- ➡ Incorrect DNS server address
- Duplicate IP addresses

EXPLANATION

In this scenario, a ping test to the website succeeds, while accessing the website through the browser does not work. Users type host names in the browser to go to websites, but host names must be translated to IP addresses by a DNS server. Either the workstation is using the wrong address for the DNS server, the DNS server is not available, or the DNS server does not have an entry for the website.

Because the ping and traceroute tests work, you know that the IP address, subnet mask, and default gateway values are correct.

REFERENCES

LabSim for Network Pro, Section 5.10.

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